

Sara Rostami

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SUMMARY

ML researcher with 4+ years building end-to-end pipelines on high-dimensional, noisy real-world data. Focused on making model outputs trustworthy — interpretability, uncertainty quantification, and calibration — across deep networks in PyTorch and LLM-based systems. Seeking Applied AI Research Scientist and Data Scientist roles.

TECHNICAL SKILLS

Languages: Python, SQL, R, MATLAB, Java, HTML/CSS
ML & Deep Learning: PyTorch, Scikit-learn, Keras — explainable AI (SHAP, LIME, D-RISE), trustworthy AI, uncertainty quantification & calibration, fine-tuning and domain adaptation, time-series forecasting, multivariate decoding (MVPA)
GenAI & LLM: Hugging Face, LangChain, LangGraph, OpenAI / Gemini APIs — RAG, agentic workflows, prompt engineering, model evaluation
Data & Infrastructure: PostgreSQL, SQL Server, Psycopg2, FastAPI, Git, pandas, NumPy, SciPy, Selenium, BeautifulSoup
Visualization: Matplotlib, Seaborn, Plotly

EXPERIENCE

Graduate Researcher — Mohsenzadeh Lab Sep 2025 – Present
University of Western Ontario & Vector Institute — London, ON, Canada

- Designed and run a simultaneous EEG/fNIRS acquisition protocol — 32-channel EEG co-registered with 32 fNIRS optodes — collecting concurrent recordings from 20+ participants across 5 image categories.
- Built the study’s end-to-end preprocessing, artifact-rejection, and feature-extraction pipeline, replacing a manual workflow with one reproducible run per subject and standardizing analysis across all 20+ datasets.
- Ran multivariate decoding over 936 trials per participant with permutation-based significance testing, linking neural signals to behavioral measures of visual category perception and image memorability.
- Designing and benchmarking neuro-inspired deep network architectures in PyTorch against experimental findings.

Research Assistant — Convergent Technologies Research Center Aug 2022 – Aug 2025
University of Tehran — Tehran, Iran

- Acquired and preprocessed 128-channel EEG data from 15 participants; applied MVPA to identify cognitive correlates of repetition and expectation effects in visual mismatch negativity.
- Built reusable univariate and multivariate workflows with permutation testing, isolating spatiotemporal signatures that informed downstream computational models.

Research Intern — Genzel Lab, Donders Institute Aug 2022 – Feb 2023
Radboud University — Nijmegen, Netherlands

- Compared 4 classifiers (logistic regression, decision tree, random forest, naive Bayes) on physiological sleep recordings; used random-forest feature importance to isolate the top predictors of memory consolidation.

SELECTED PROJECTS

Calibrated Uncertainty in Vision-Language Models — Post-hoc calibration of a frozen BLIP-2 on VQA v2: compared temperature scaling, MC Dropout, and Laplace approximation (ECE, NLL) for confidence-based abstention; temperature scaling cut ECE from 0.321 to 0.097. [GitHub](#)
Neuro-Sentry — Hack Western 12 — Real-time multimodal stroke-risk triage: a SwiftUI iOS client streams facial-mesh landmarks and vitals to a FastAPI backend, where a Gemini pipeline grounded in clinical guidelines returns structured risk scores to a React dashboard. [Devpost](#)
Explainable AI: SHAP, LIME, and D-RISE — SHAP (Deep/Kernel) on a WHO life-expectancy network identifying schooling as the dominant predictor; D-RISE for model-agnostic object-detection explanations; LIME superpixel saliency on MobileNetV2. [GitHub](#)
Additional projects — GANs (DCGAN/AC-GAN/WGAN-GP), BERT & BEiT attention implementations, fraud detection and OCR with deep networks, cohort-based customer retention analysis, audio classification. github.com/SaraRostami/University

EDUCATION

MSc Neuroscience (Machine Learning Specialization), University of Western Ontario — GPA 3.97/4.0 Sep 2025 – Aug 2027
MSc Artificial Intelligence and Robotics, University of Tehran — GPA 19.12/20 (≈3.9/4.0); ranked 3rd in cohort Sep 2021 – 2024
BSc Computer Engineering, Babol Noshirvani University of Technology Sep 2016 – 2021

AWARDS & TEACHING

Awards: Western University Graduate Funding Package (2025–2027) · Vector Institute Research Grant (2025–2026).
Teaching: Neuromatch Academy Comp. Neuroscience Summer School (2023, 2025, 2026), project-pod TA; Western University (2026), Python and Java programming; University of Tehran (2023), Statistical Inference and Cognitive Neuroscience.